Dimension Reduction: Manifold Learning, Data Representation and A Unified Framework

COMP7404

Computational Intelligence and Machine Learning

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Outline

- ✓Manifold Learning Methods
- ✓Dimensionality Reduction for Tensor-based Objects
- ✓Graph Embedding: A General Framework for Dimensionality Reduction

LLE

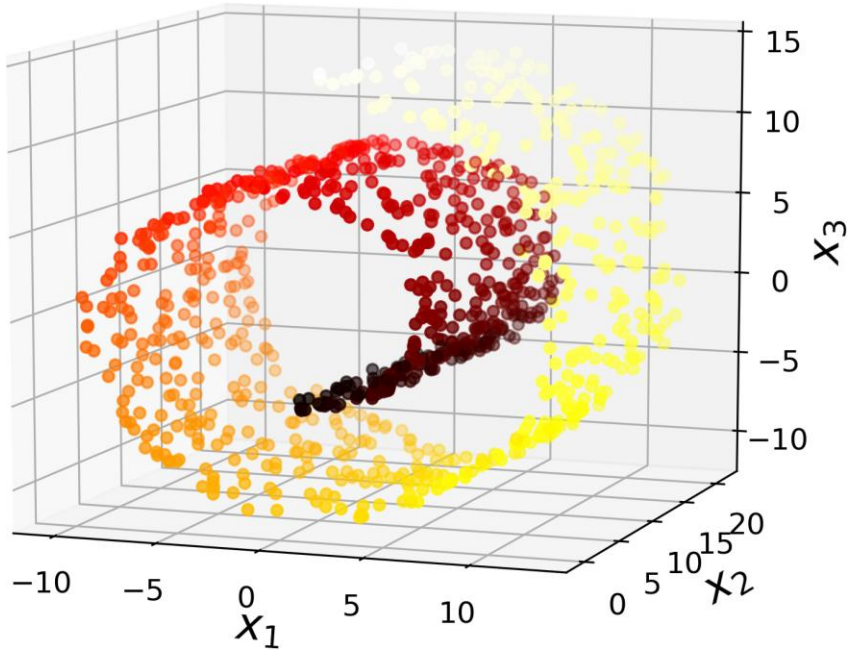
- How it works
  - Measures how each training instance linearly relates to its closest neighbors
  - Then looks for a low-dimensional representation of the training set where these local relationships are best preserved
- This approach makes it particularly good at unrolling twisted manifolds, especially when there is not too much noise

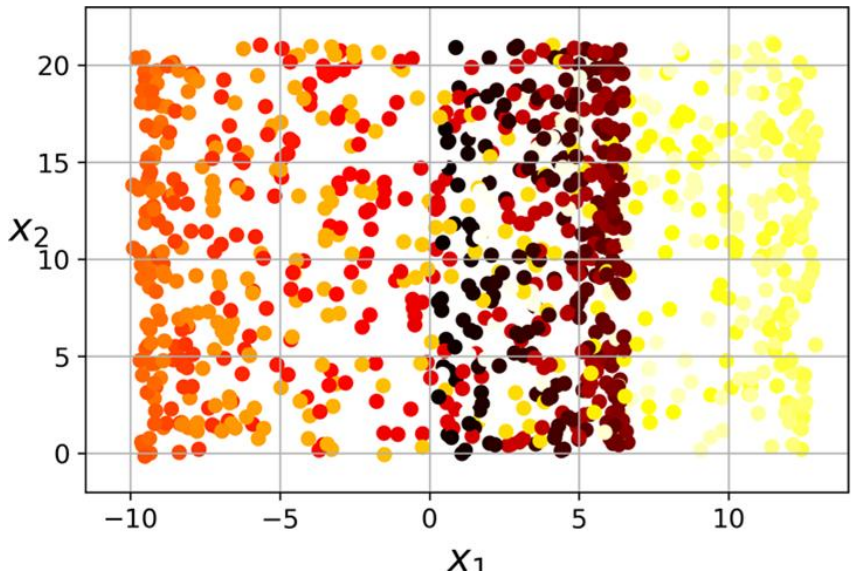
```
1 from sklearn.manifold import LocallyLinearEmbedding
2 lle = LocallyLinearEmbedding(n_components=2, n_neighbors=10)
3 X_reduced = lle.fit_transform(X)
```

- Available here on CoLab

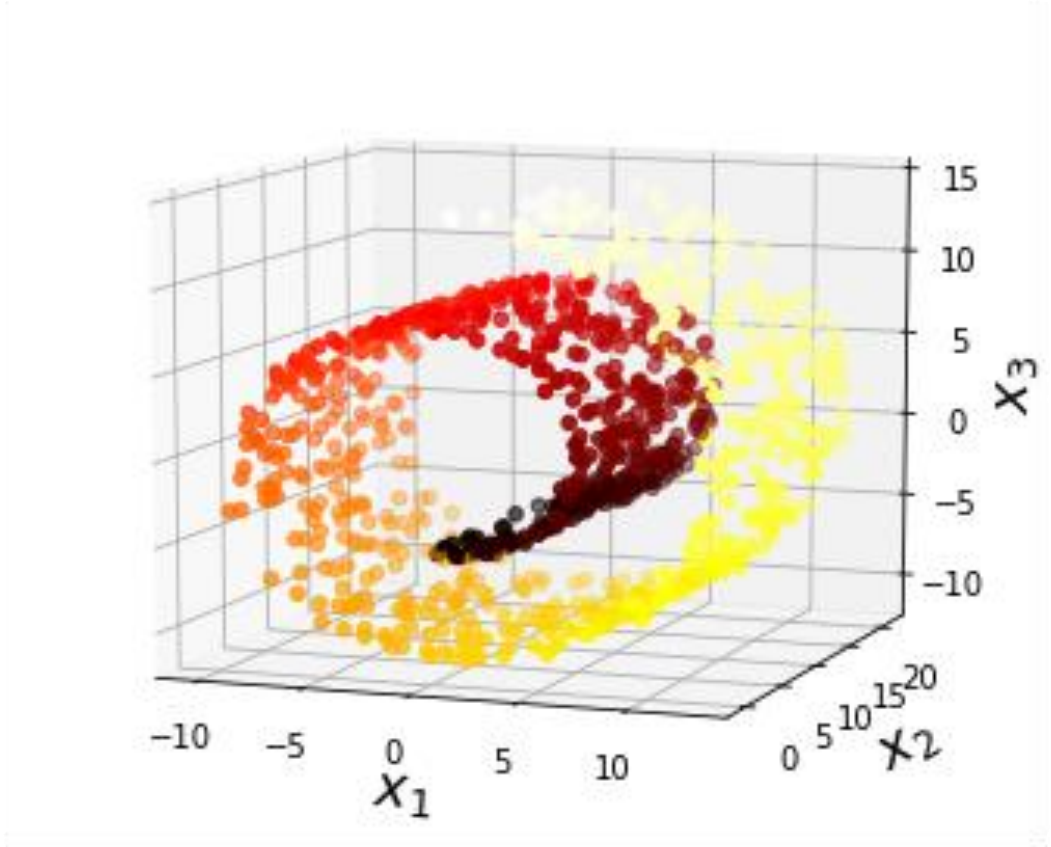
When Projection Fails

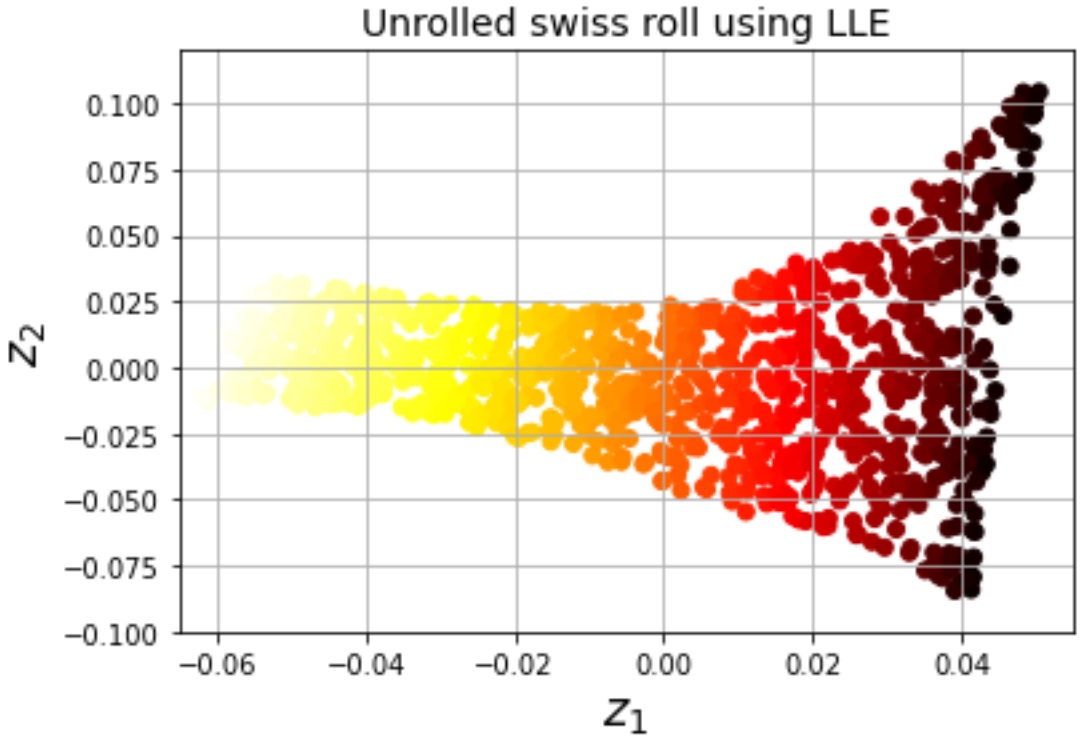
- Projection is not always the best approach
- Consider the following toy dataset to illustrate this problem
  - The Swiss roll
- Simply projecting onto a plane (e.g., dropping x3) would squash different layers



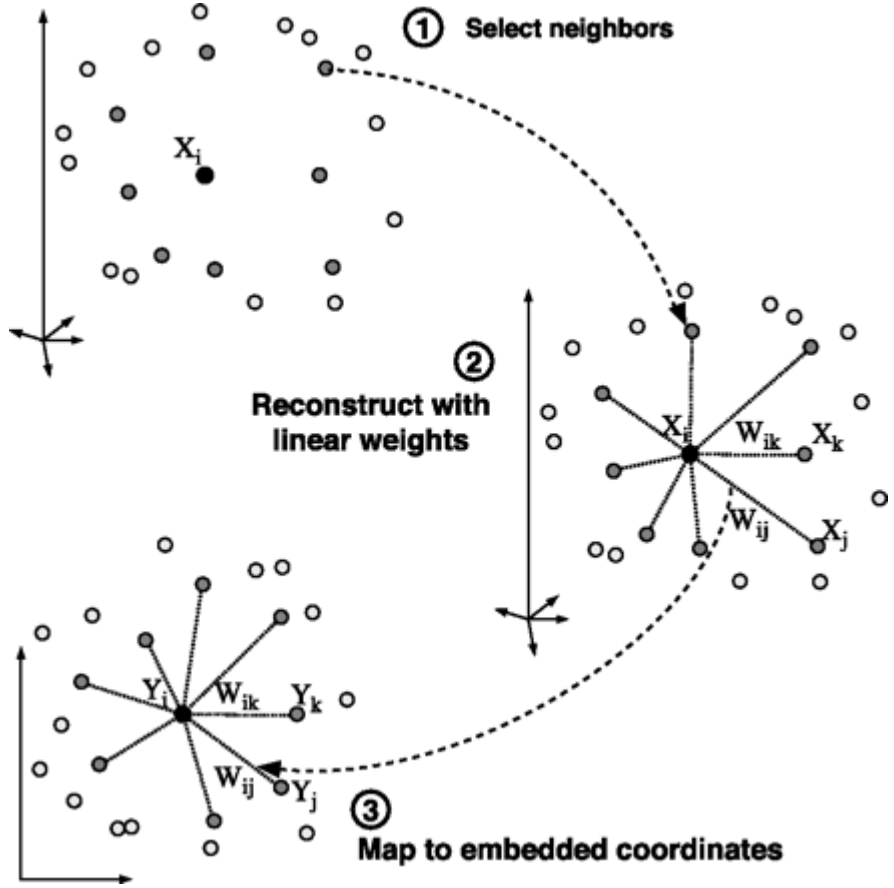


LLE: Unrolling the Swiss roll





Locally Linear Embedding (LLE)



- History: S. Roweis and L. Saul, Science, 2000
- Procedure
  1. Identify the neighbors of each data point
  2. Compute weights that best linearly reconstruct the point from its neighbors

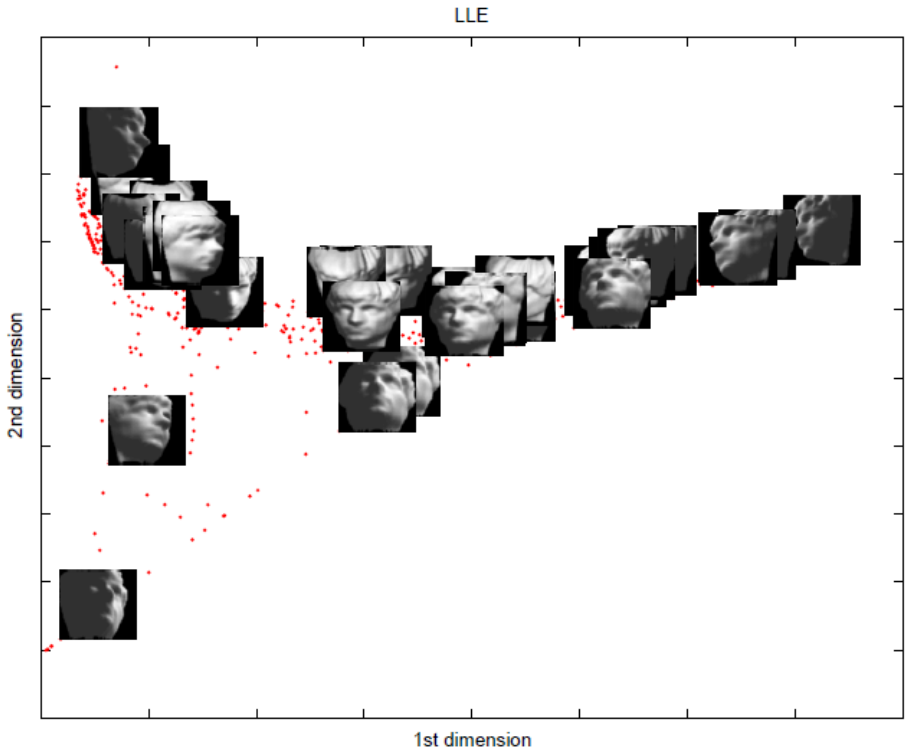
$$\min_{\mathbf{w}} \sum_{i=1}^{N} \|\mathbf{x}_i - \sum_{j=1}^{k} w_{ij}\mathbf{x}_{N_i(j)}\|^2$$

3. Find the low-dimensional embedding vector which is best reconstructed by the weights determined in Step 2

$$\min_{Y} \sum_{i=1}^{N} \|\mathbf{y}_i - \sum_{j=1}^{k} w_{ij}\mathbf{y}_{N_i(j)}\|^2 \Leftrightarrow \min_{Y} \operatorname{tr}(Y^\top Y L)$$

Centering Y with unit variance

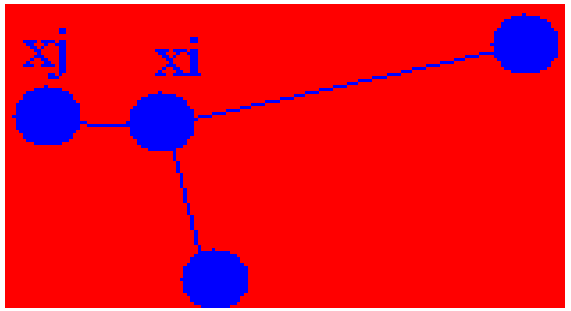
LLE Example



Laplacian Eigenmaps (LEM)

- History: M. Belkin and P. Niyogi, 2003
- Similar to locally linear embedding
- Different in weights setting and objective function
  - Weights

$$W_{ij} = \begin{cases} 1 & i,j \text{ are connected} \\ \exp\left(\frac{-\|x_i - x_j\|^2}{s}\right) & \text{otherwise} \end{cases}$$

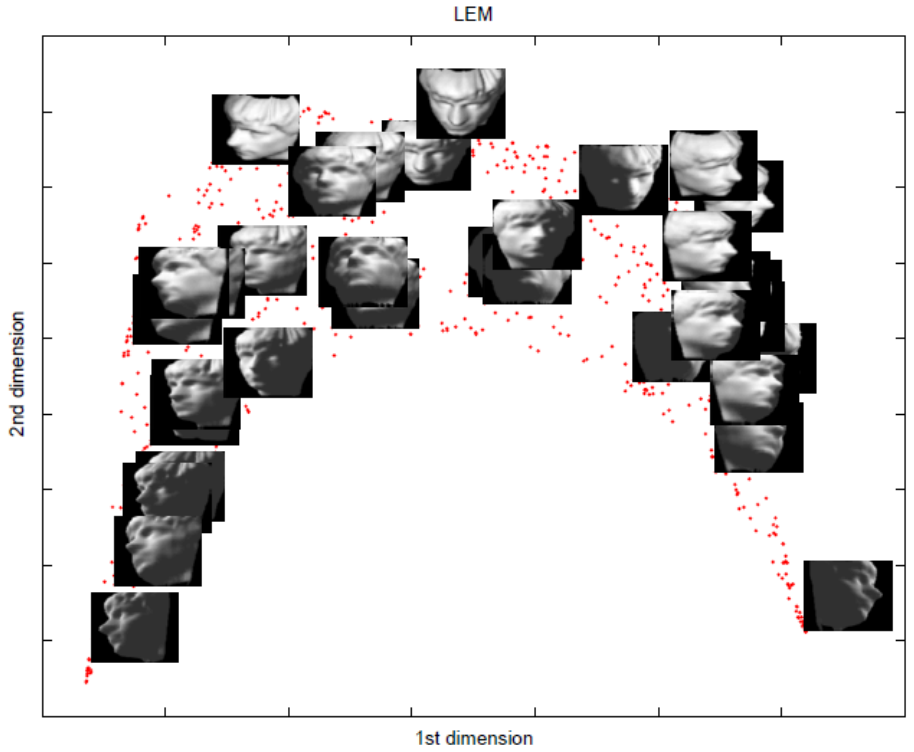


  - Objective

$$\min_Y \sum_{i=1}^{N}\sum_{j=1}^{N}(\mathbf{y}_i - \mathbf{y}_j)W_{ij} \iff \min_Y \operatorname{tr}(YLY^\top)$$

where $L = R - W$, R is diagonal and $R_{ii} = \sum_{j=1}^{N} W_{ij}$.

LEM Example



Multidimensional Scaling (MDS)

- History: T. Cox and M. Cox, 2001
- Attempts to preserve pairwise distances
- Different formulation of PCA, but yields similar result form

$$\min_{Y} \sum_{i=1}^{N}\sum_{j=1}^{N}(d_{ij}^{(X)} - d_{ij}^{(Y)})^2$$

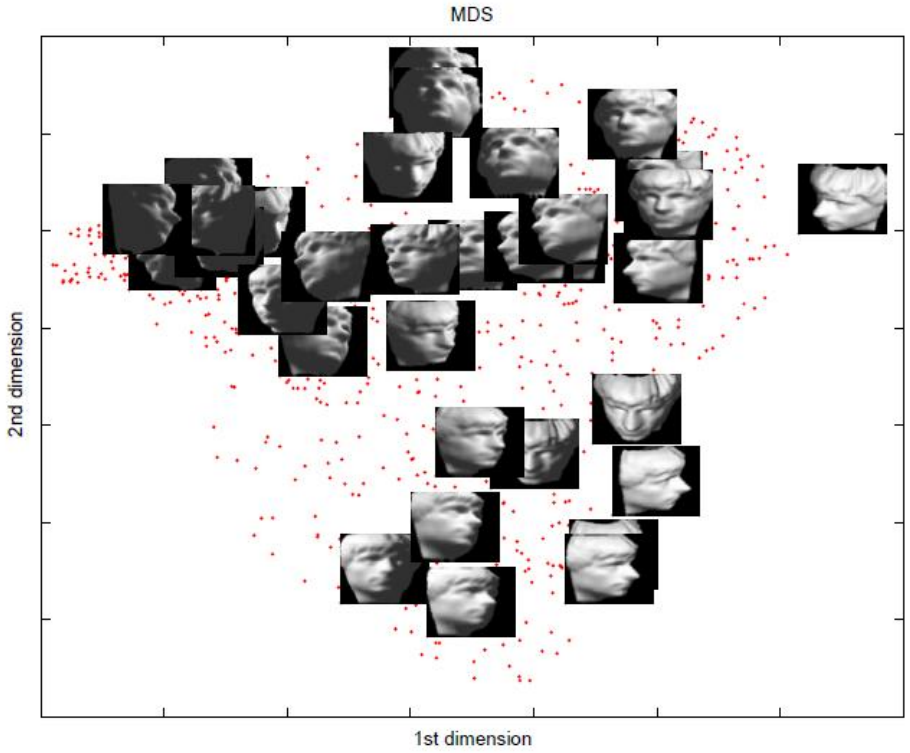
$$\text{where } d_{ij}^{(X)} = \|x_i - x_j\|^2 \text{ and } d_{ij}^{(Y)} = \|y_i - y_j\|^2.$$

- Transformation

$$X^\top X = -\frac{1}{2} H D^{(X)} H \qquad \text{where } H = I - \tfrac{1}{N}\mathbf{1}\mathbf{1}^\top.$$

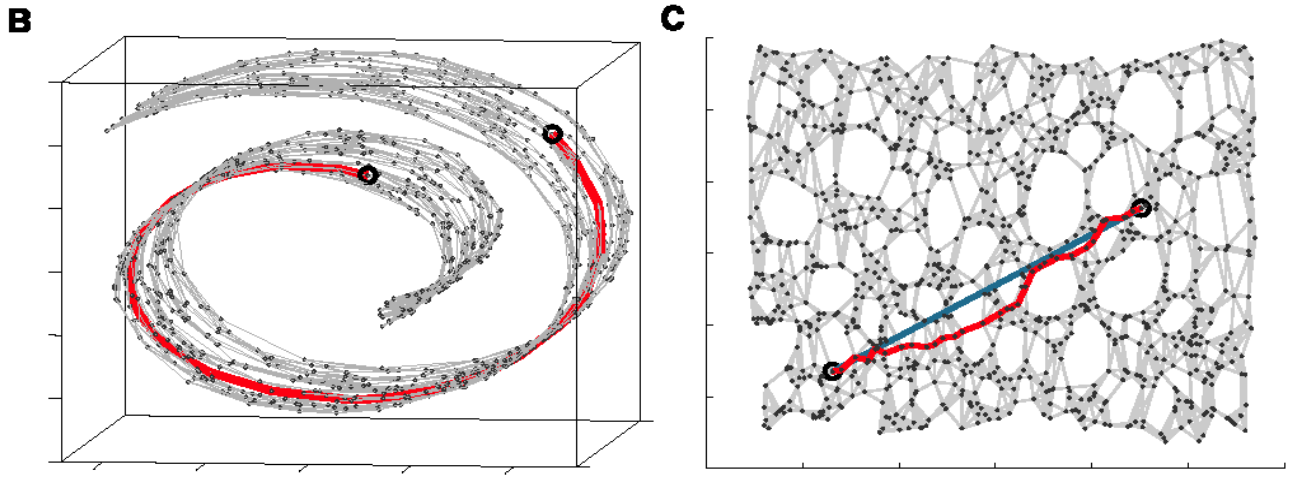
$$\min_{Y} \sum_{i=1}^{N}\sum_{j=1}^{N}(x_i^\top x_j - y_i^\top y_j)^2$$

MDS Example



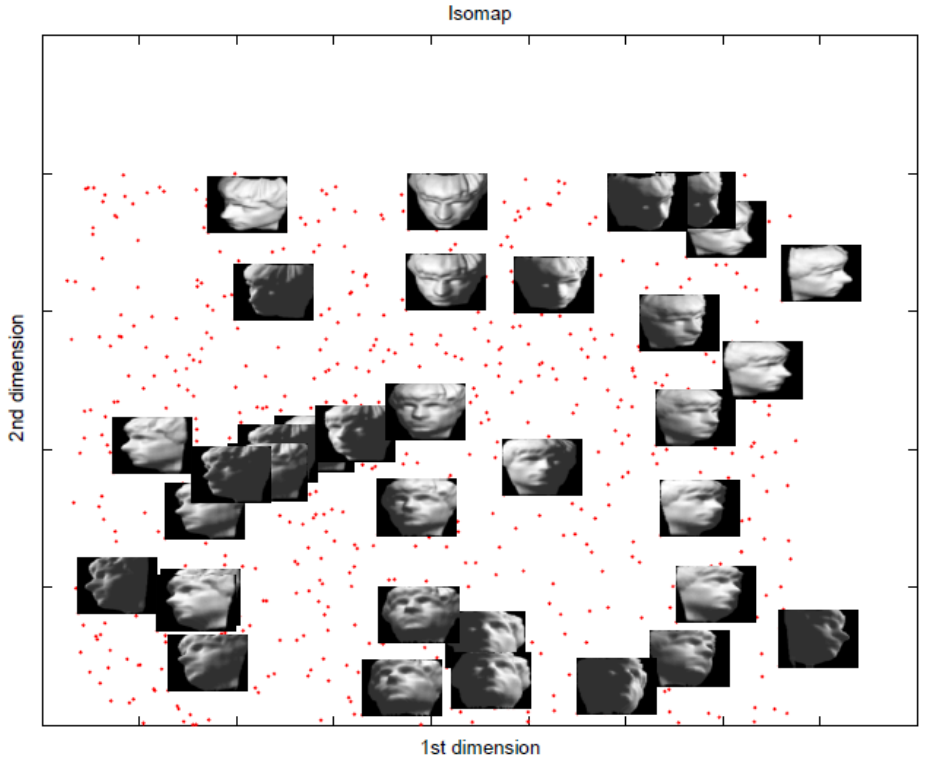
Isomap

- History: J. Tenenbaum et al, Science 1998
- A nonlinear generalization of classical MDS
- Perform MDS, not in the original space, but in the geodesic space
- Procedure-similar to LLE
  1. Find neighbors of each data point
  2. Compute geodesic pairwise distances (e.g., shortest path distance) between all points
  3. Embed the data via MDS



Example: 3D space to 2D space

Isomap Example



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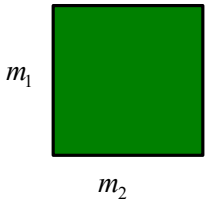
What is Tensor?

Tensors are arrays of numbers which transform in certain ways under coordinate transformations.

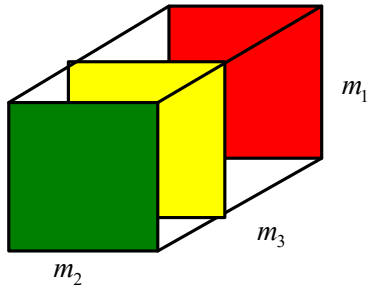
$x \in \mathrm{R}^{m_1}$

Vector



$X \in \mathrm{R}^{m_1 \times m_2}$

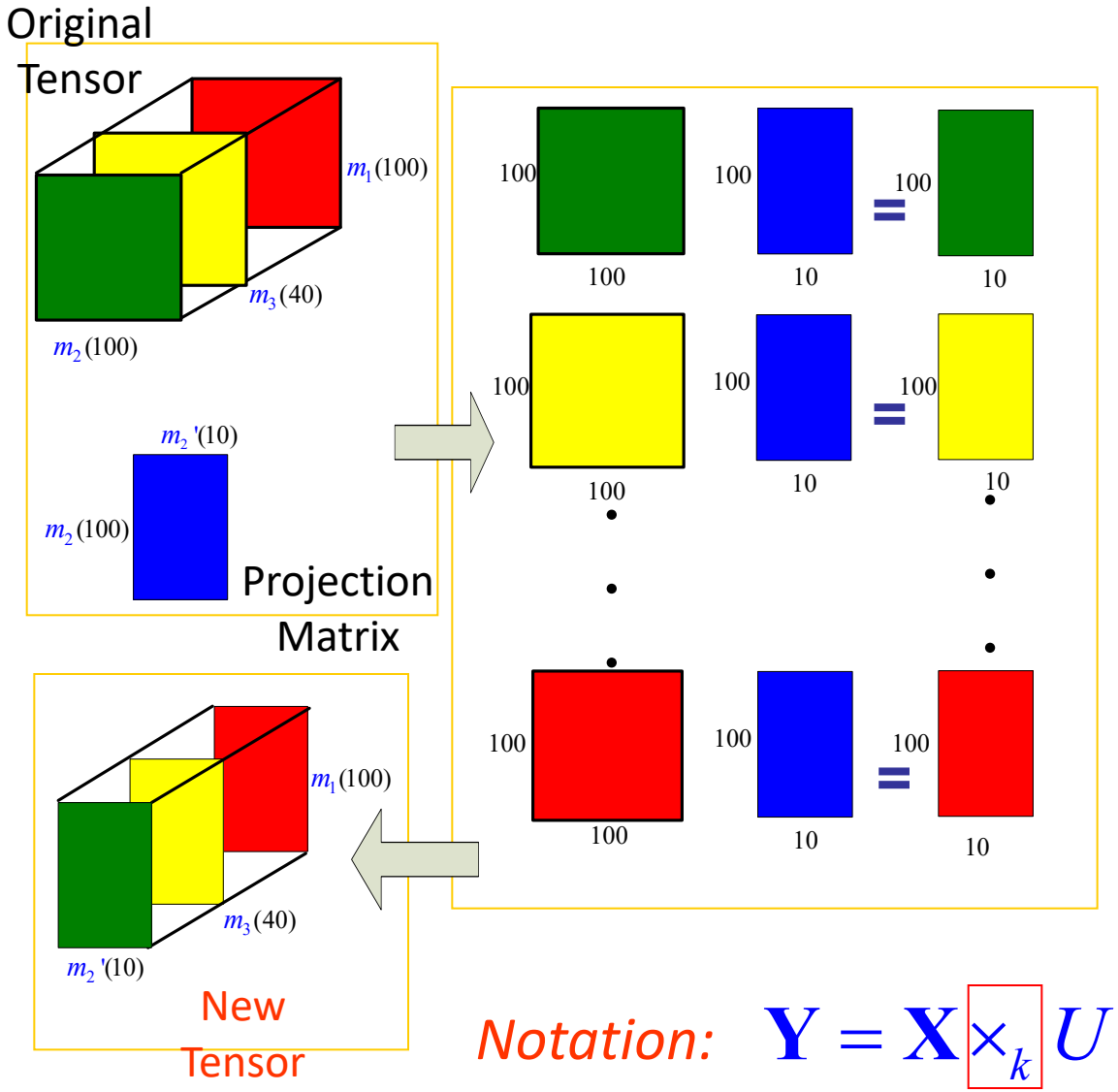
Matrix



$\mathbf{X} \in \mathrm{R}^{m_1 \times m_2 \times m_3}$

3rd-order Tensor

Definition of Mode-k Product



Notation: $\mathbf{Y} = \mathbf{X} \times_k U$

Projection:

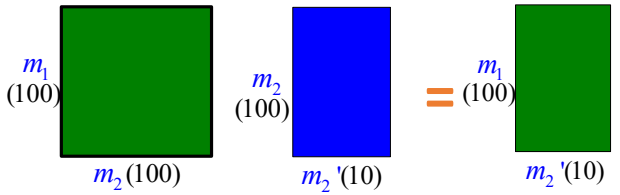
high-dimensional space
-> low-dimensional space

Reconstruction:

low-dimensional space
-> high-dimensional space

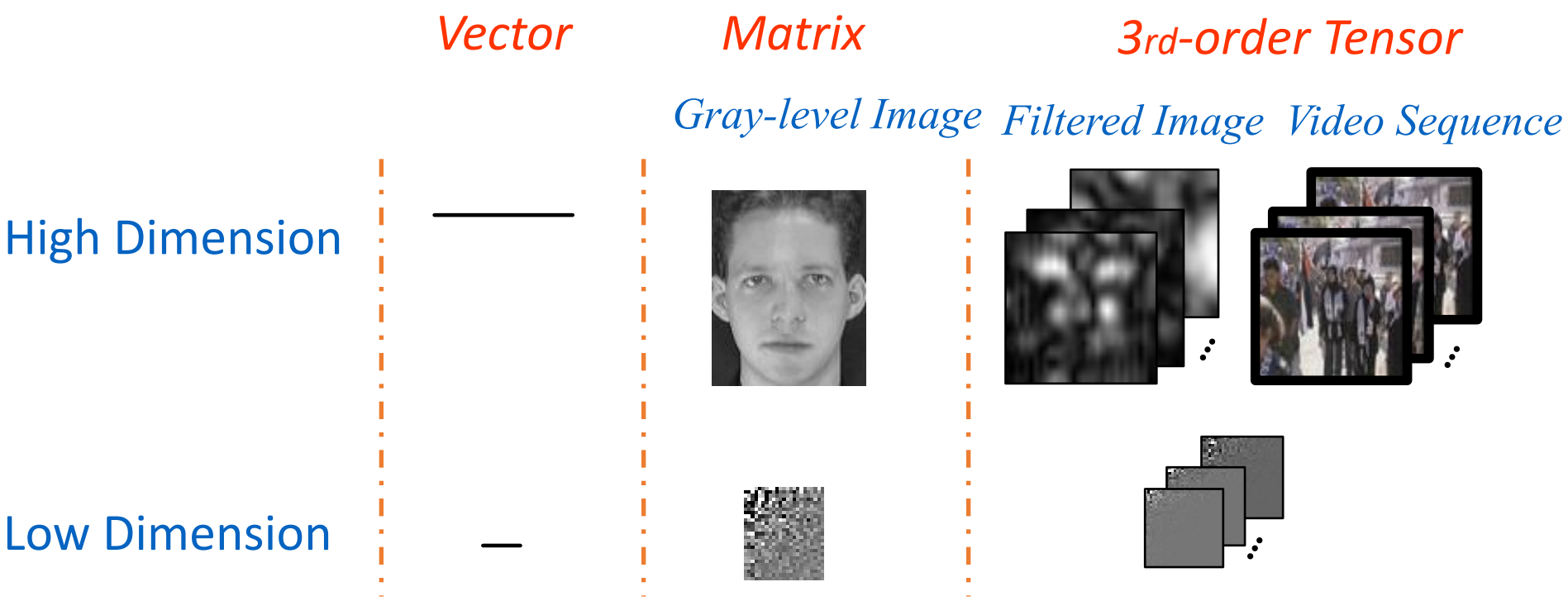
Product for two Matrices

$$Y = XU \quad Y_{ij} = \sum_{k=1}^{m_2} X_{ik} U_{kj}$$



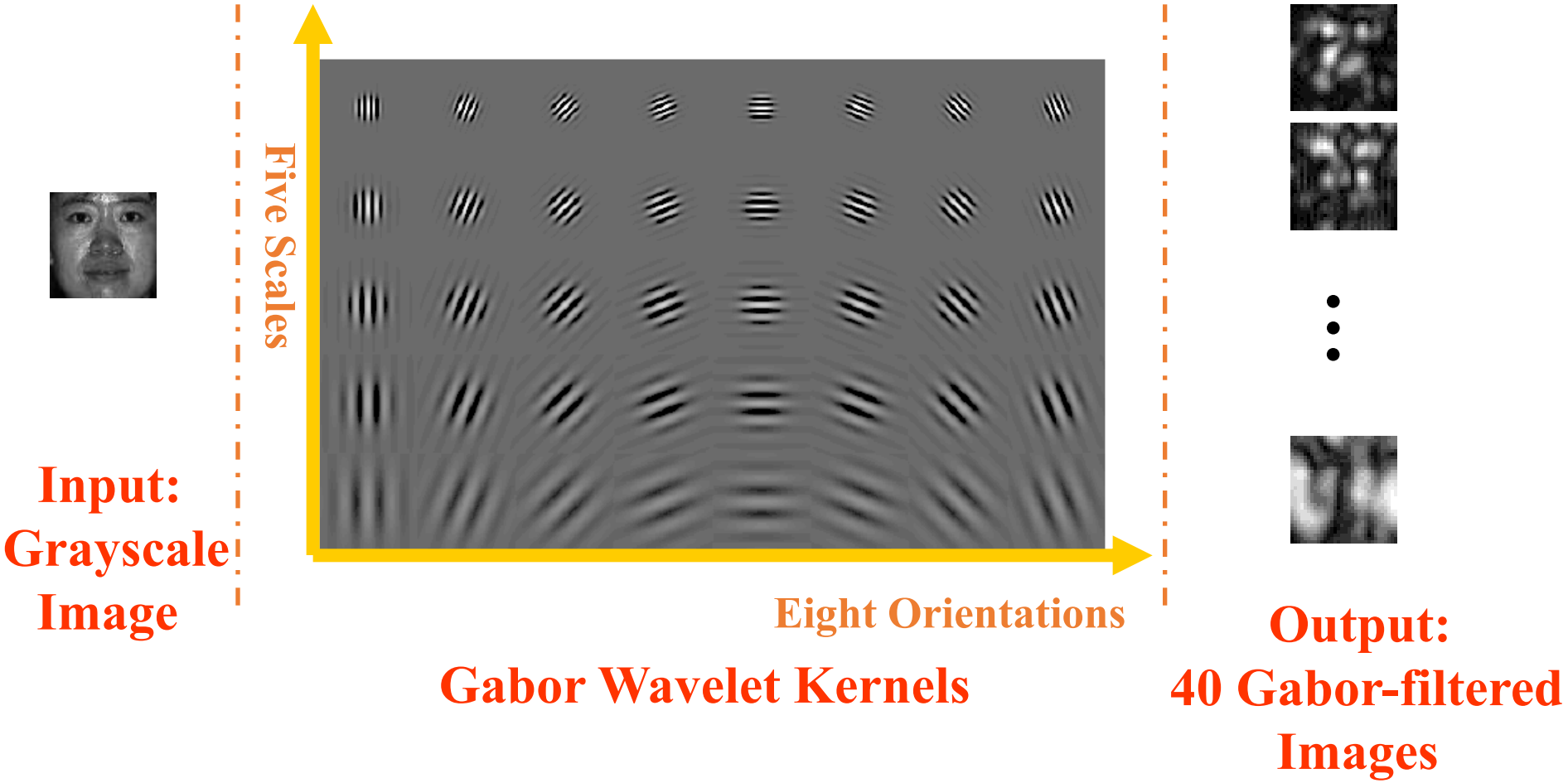
Original Matrix | Projection Matrix | New Matrix

Data Representation in Dimensionality Reduction



What is Gabor Features?

Gabor features can improve recognition performance in comparison to grayscale features. *Chengjun Liu T-IP, 2002*



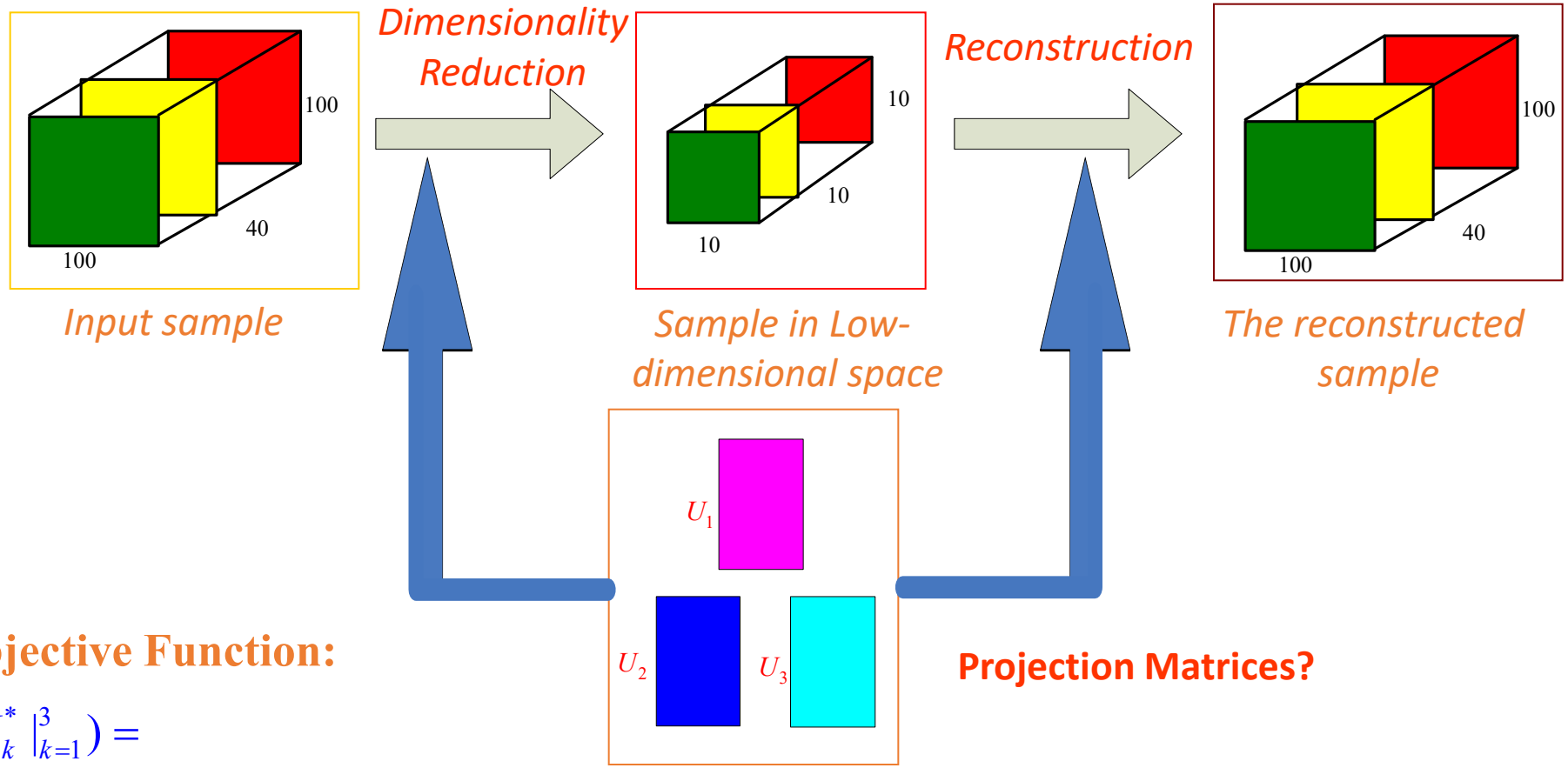
Why Represent Objects as *Tensors* instead of Vectors?

- *Natural Representation*
  - *Gray-level Images (2D structure)*
  - *Videos (3D structure)*
  - *Gabor-filtered Images (3D structure)*

- *Enhance Learnability in Real Application*
  - *Curse of Dimensionality* (*Gabor-filtered image*: 100*100*40 -> *Vector*: *400,000*)
  - *Small sample size problem (less than 5,000 images in common face databases)*

- *Reduce Computation Cost*

Concurrent Subspace Analysis as an Example (Criterion: *Optimal Reconstruction*)



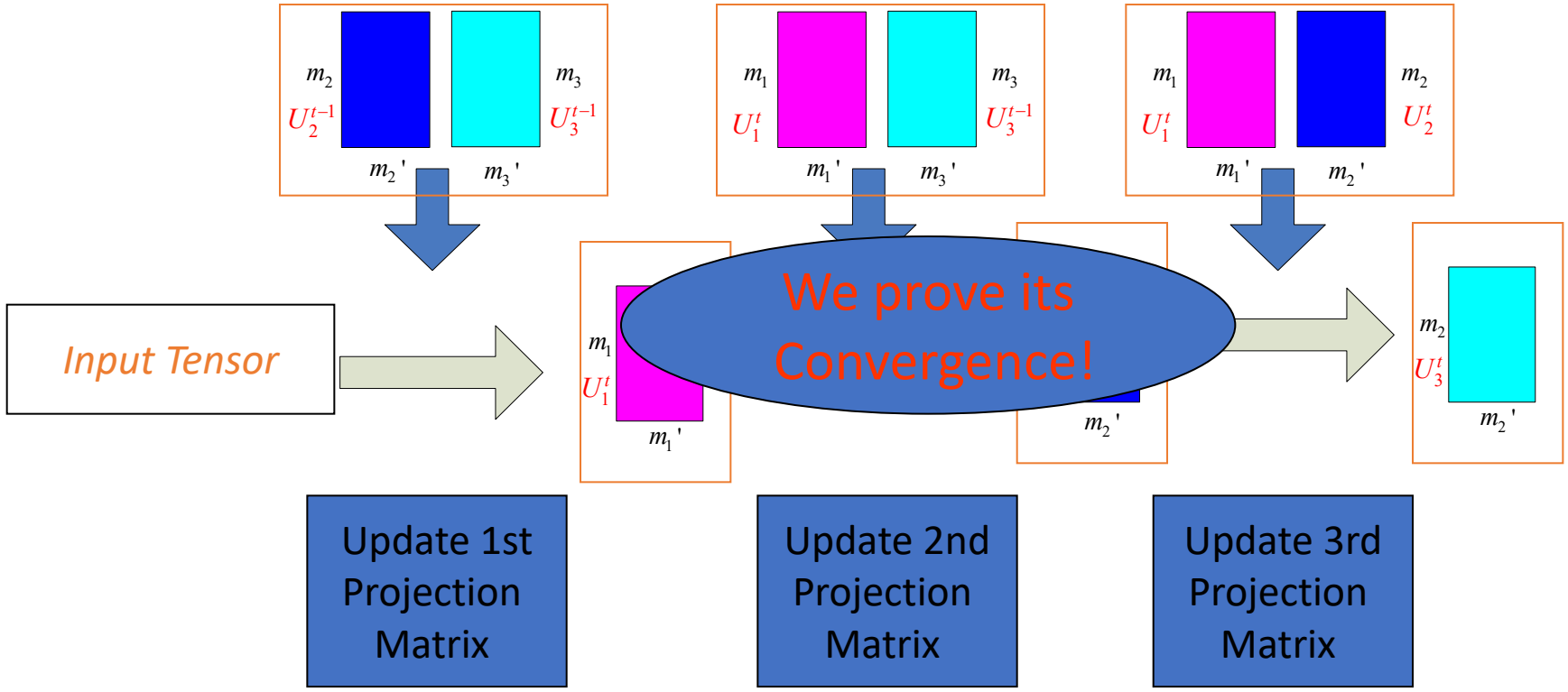
Objective Function:

$$(U_k^* \mid_{k=1}^3) = \arg\min_{U_k \mid_{k=1}^3} \sum_i \| \mathbf{X}_i \times_1 U_1 U_1' \ldots \times_3 U_3 U_3' - \mathbf{X}_i \|^2$$

Projection Matrices?

D. Xu, S. Yan, H. Zhang and et al., *CVPR, 2005*

Iterative Solution



Variants of Rank-(R1,R2,…,Rn) Approximation

L. Lathauwer, B. Moor and J. Vandewalle, SIAM Journal of Matrix Analysis and Applications, 2001

Tensor Representation: *Advantages*

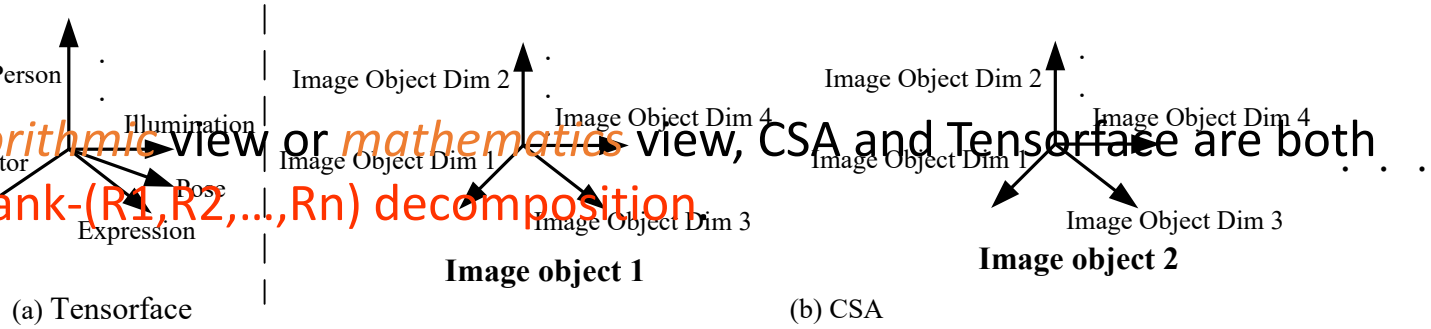
1. *Enhanced learnability*
2. *Appreciable reductions in computational costs*
3. *Large number of available projection directions*
4. *Utilize the structure information*

Correlation along the column vectors of the *mode-k* flattened matrices.

	PCA	CSA
Feature Dimension	$\prod_{k=1}^{n} m_k$	m_k
Sample Number	N	$\prod_{i \neq k} m_i * N$
Computation Complexity	$O(m^{3n})$	$O(n * m^{n+1})$

Connection to Previous Work –*Tensorface*

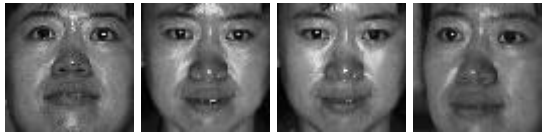
(M. Vasilescu and D. Terzopoulos, 2002)



From an *algorithmic* view or *mathematics* view, CSA and Tensorface are both variants of Rank-(R1,R2,…,Rn) decomposition.

	Tensorface	CSA
Motivation	Characterize *external* factors	Characterize internal factors
Input: Gray-level Image	*Vector*	Matrix
Input: Gabor-filtered Image (Video Sequence)	*Not address*	3rd-order tensor
When equal to PCA	The number of images per person are only *one* or are *a prime number*	Never
Number of Images per Person for Training	*Lots of images* per person	One image per person

Experiments: *Database Description*

	Number of Persons (Images per person)	Image Size (Pixels)	Example Images
Simulated Video Sequence	60 (1)	64×64×13	
ORL database	40 (10)	56×46	
CMU PIE-1 sub-database	60 (10)	64×64	
CMU PIE-2 sub-databases	60 (10)	64×64	

Experiments: *Object Reconstruction (1)*

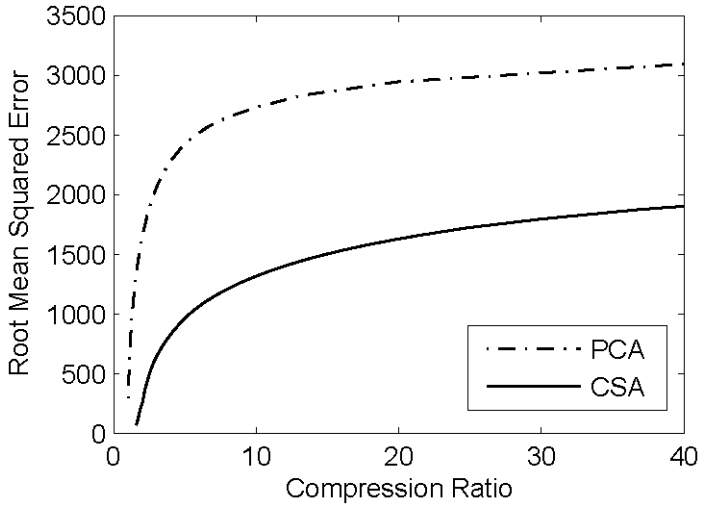
Input: Gabor-filtered images

ORL database

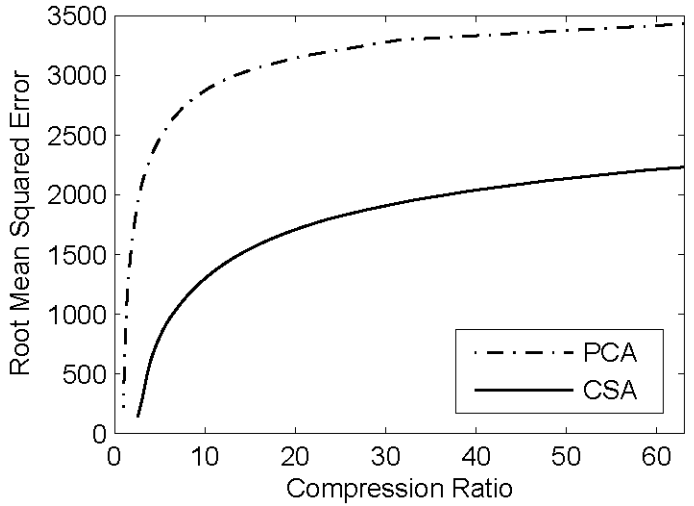
CMU PIE-1 database

Objective Evaluation Criterion:

Root Mean Squared Error (**RMSE**) and Compression Ratio (**CR**)



ORL database



CMU PIE-1 database

Experiments: *Object Reconstruction (2)*

Input: Simulated video sequence

Original Images

Reconstructed Images from PCA

Reconstructed Images from CSA

Experiments: *Face Recognition*

Input: Gray-level images and Gabor-filtered images
ORL database
CMU PIE database

Algorithm	CMU PIE-1	CMU PIE-2	ORL
PCA (Gray-level feature)	70.1%	28.3%	76.9%
PCA (Gabor feature)	80.1%	42.0%	86.6%
CSA (Ours)	***90.5%***	***59.4 %***	***94.4%***

Summary

- This is the first work to address dimensionality reduction with a tensor representation of arbitrary order.
- Opens a new research direction.

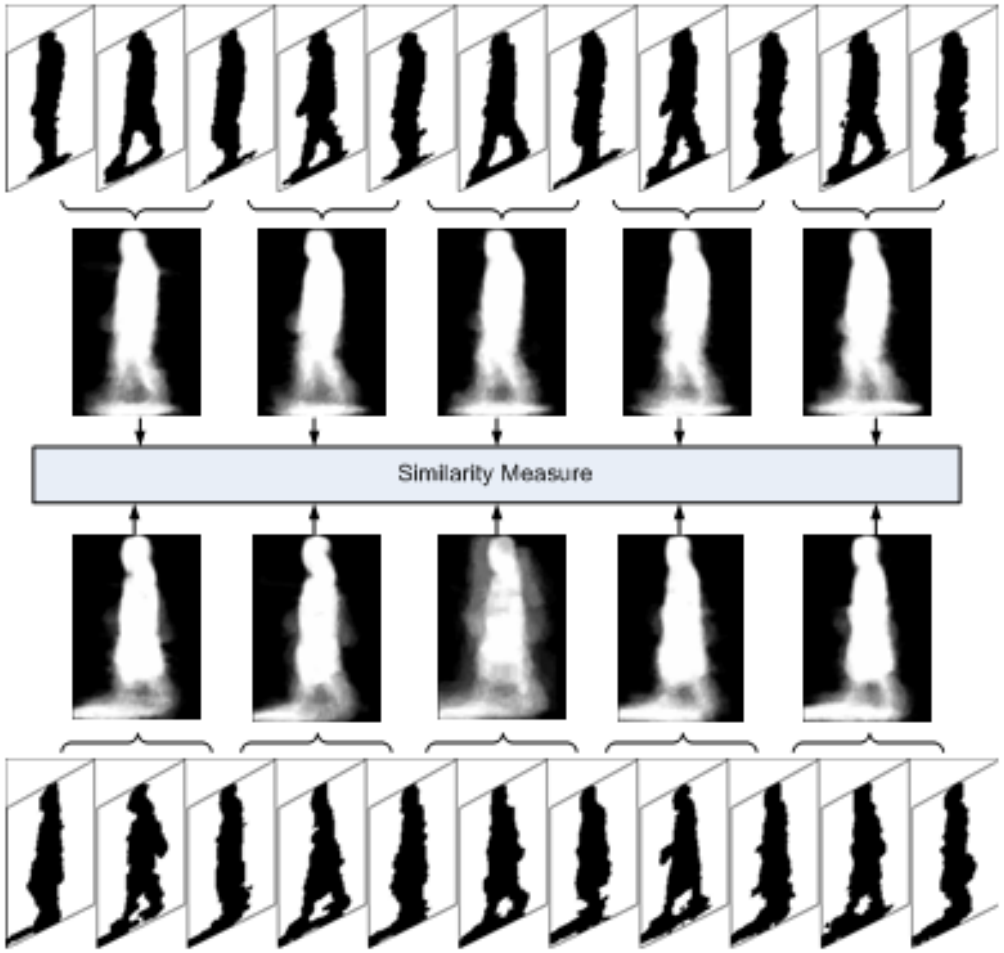
Tensorization - *New* Research Direction

- Our Extensions:
  1) Supervised Learning with Rank-(R1, R2,… Rn) Decomposition (DATER):
     CVPR 2005 and T-IP 2007
  2) Supervised Learning with Rank-1 Decomposition and Adaptive Margin (RPAM):
     CVPR 2006 and T-SMC-B 2007
  3) Application in Human Gait Recognition (CSA-2+DATER-2): *T-CSVT 2006 and T-IP 2007*
  4) Element Rearrangement: *CVPR 2007 and T-PAMI 2009*

- D. Tao, S. Maybank, et al.'s Extensions :
  1)Incremental Learning with Tensor Representation: *ACM SIGKDD 2006*
  2)Tensorized *SVM* and *Minimax Probability Machines*: *ICDM 2005*

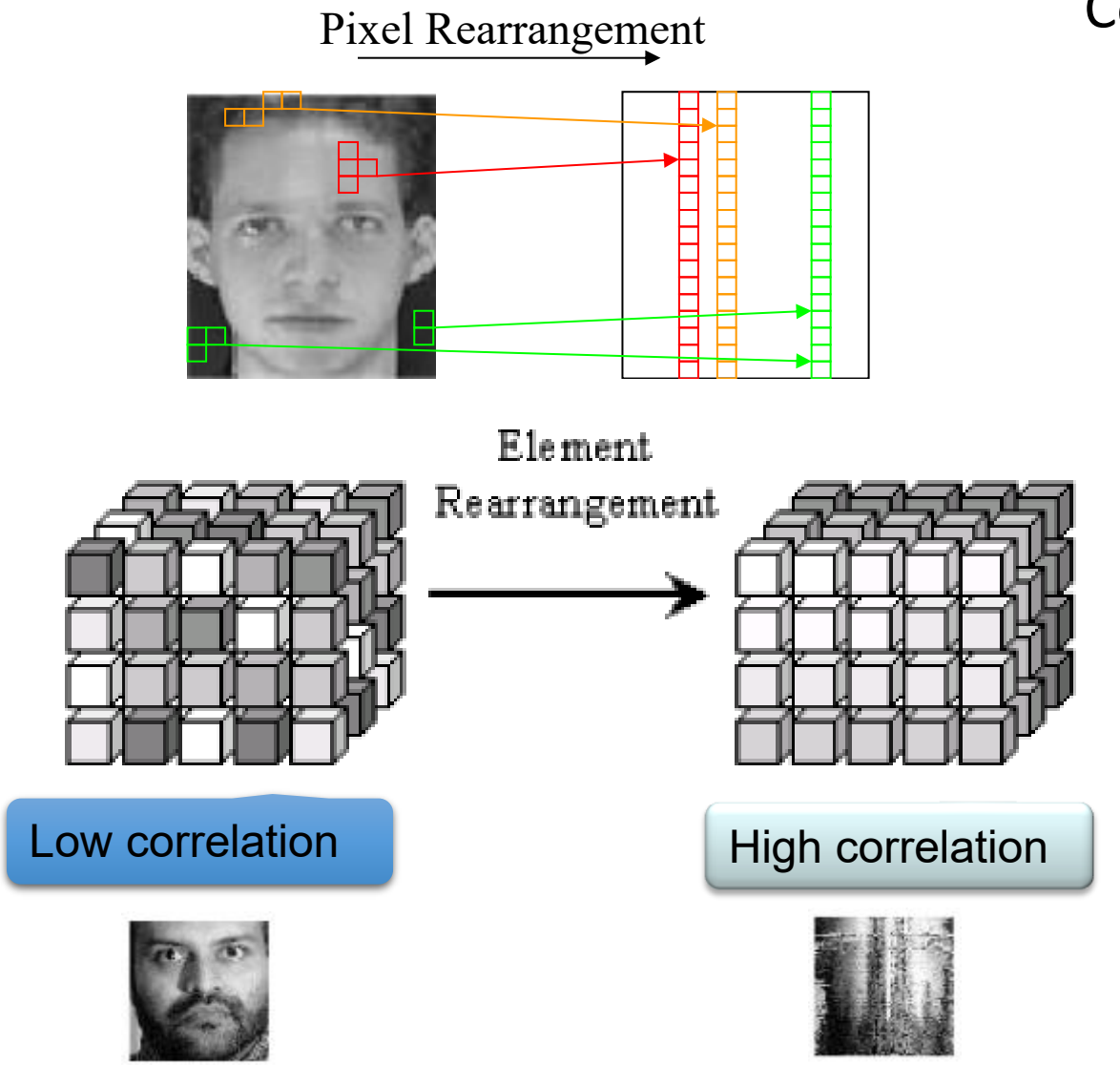
- G. Dai and D. Yeung's Extensions:
  Tensorized *NPE* (Neighborhood Preserving Embedding), *LPP* (Locality Preserving Projections) and *LDE* (Local Discriminant Embedding): *AAAI 2006*

Human Gait Recognition with *Matrix Representation*



D. Xu, S. Yan, H. Zhang and et al., *T-CSVT, 2006 and T-IP 2007*

Element Rearrangement (CVPR07 and T-PAMI09)



Contradiction:

- The success of tensor-based subspace learning relies on the redundancy among the unfolded vector
- The truth is that this kind of correlation/redundancy is often not strong in real data

Problem Definition

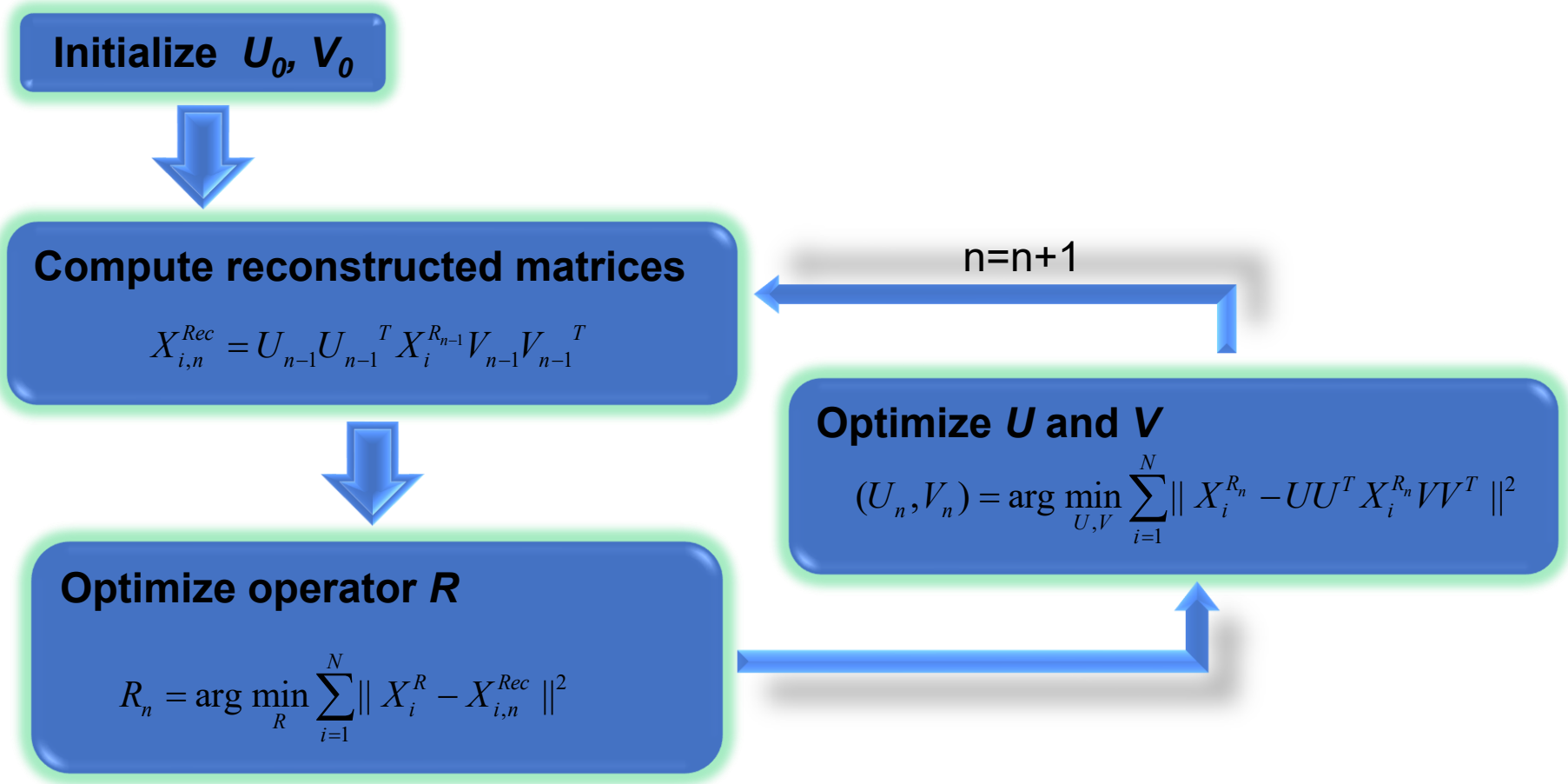
- The task of enhancing correlation/redundancy among 2^{nd}–order tensor is to search for a pixel rearrangement operator R, such that

$$R^* = \arg\min_R \{ \min_{U,V} \sum_{i=1}^{N} \| X_i^R - UU^T X_i^R VV^T \|^2 \}$$

1. X_i^R is the rearranged matrix from sample X_i
2. The column numbers of U and V are predefined

After the pixel rearrangement, we can use the rearranged tensors as input for Tensorization of graph embedding!

Solution to Pixel Rearrangement Problem



$$Note: \sum_{i=1}^{N} \| X_i^{R_n} - X_{i,n}^{Rec} \|^2 \geq \sum_{i=1}^{N} \| X_i^{R_n} - U_{n-1}U_{n-1}^{T}X_i^{R_n}V_{n-1}V_{n-1}^{T} \|^2$$

Step for Optimizing R

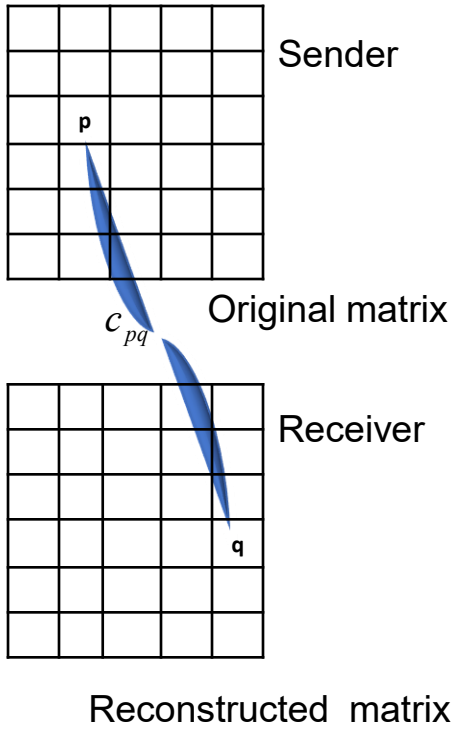
- It is Earth Mover Distance problem

$$R^* = \arg\min_R \sum_{i=1}^{N} || X_i^R - X_{i,n}^{Rec} ||^2$$

$$\min_R \sum_{p,q} c_{pq} R_{pq} \quad st.$$

$$1: 0 \le R_{pq} \le 1; \quad 2: \sum_p R_{pq} = 1; \; 3: \sum_q R_{pq} = 1$$

$$where \; c_{pq} = \sum_{i=1}^{N} | X_i(p) - X_{i,n}^{Rec}(q) |^2$$



1. Linear programming problem has integer solution.
2. We constrain the rearrangement within local neighborhood for speedup.

Convergence Speed

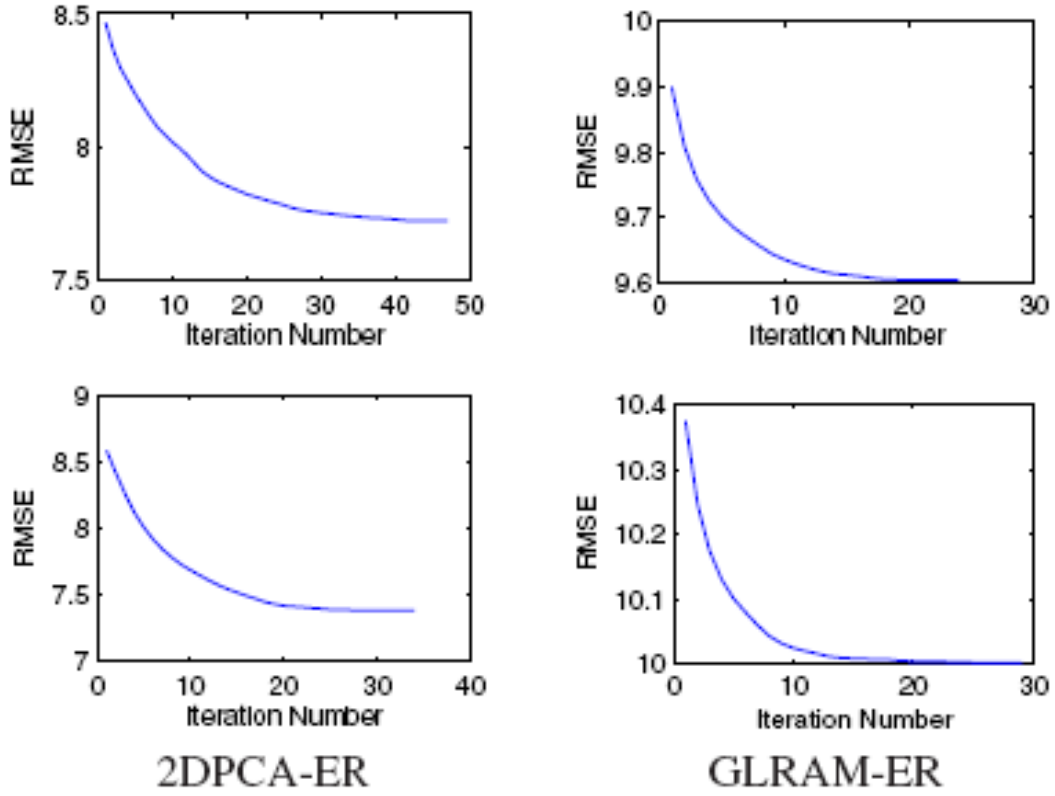


Figure 2. Algorithm convergence with element rearrangement. Root Mean Squared Error (RMSE) vs. number of iterations on the CMU PIE (top) and FERET (bottom) databases.

Rearrangement Results

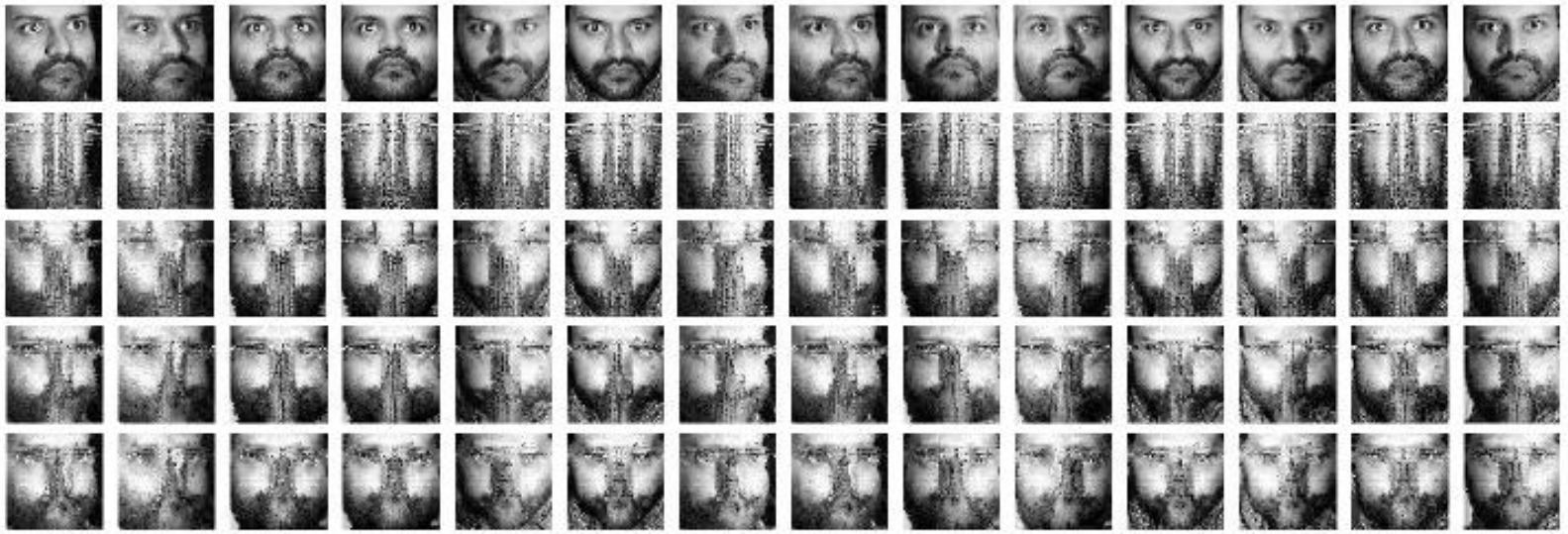


Figure 3. Comparison of fourteen element-rearranged images from the CMU PIE database based on the GLRAM-ER algorithm. The top row displays the original images. The subsequent rows show the rearranged images for $d = 2, 3, 4$, and 5, respectively.

Reconstruction Visualization

Figure 5. Comparison of seven reconstructed images from the FERET database. The top row displays the original images. The subsequent rows show reconstructed results from 2DPCA, 2DPCA-ER, GLRAM, and GLRAM-ER, respectively.

Reconstruction Visualization

Figure 6. Comparison of fourteen reconstructed images from the CMU PIE database. The top row displays the original images. The subsequent rows show reconstructed results from 2DPCA, 2DPCA-ER, GLRAM, and GLRAM-ER, respectively.

Classification Accuracy

Algorithm	FERET	CMU PIE
PCA	45.1	43.3
LDA	91.8	76.2
MFA	92.1	77.3
2DLDA	94.8	81.9
2DMFA	94.0	82.8
2DLDA-PA	95.6	83.7
2DMFA-PA	96.5	86.4

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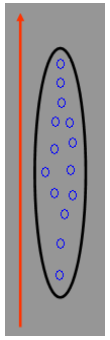
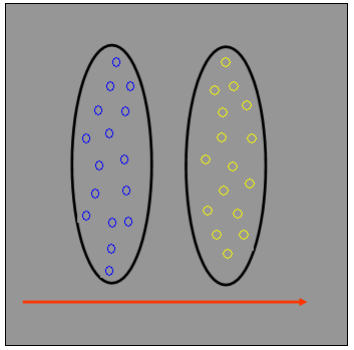
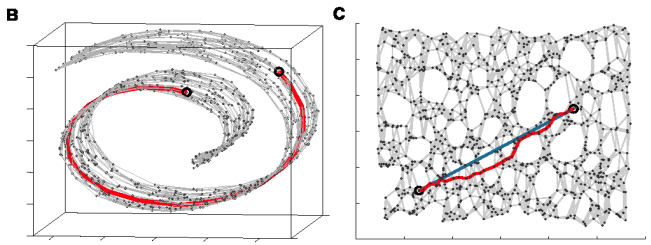
Representative Previous Work

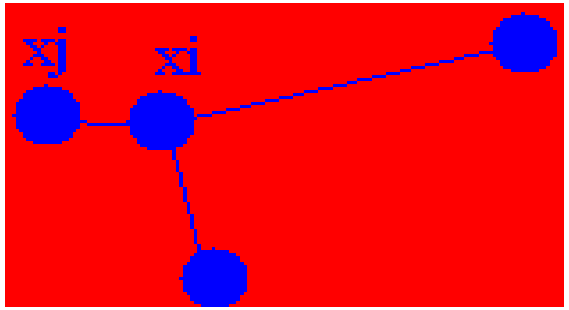


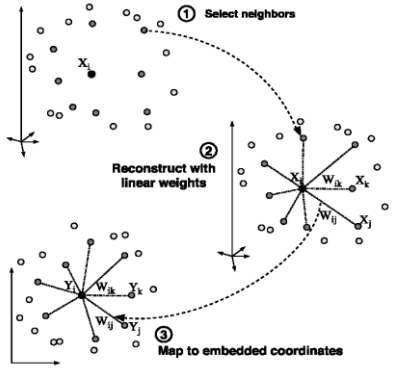


PCA

LDA







ISOMAP: Geodesic Distance Preserving

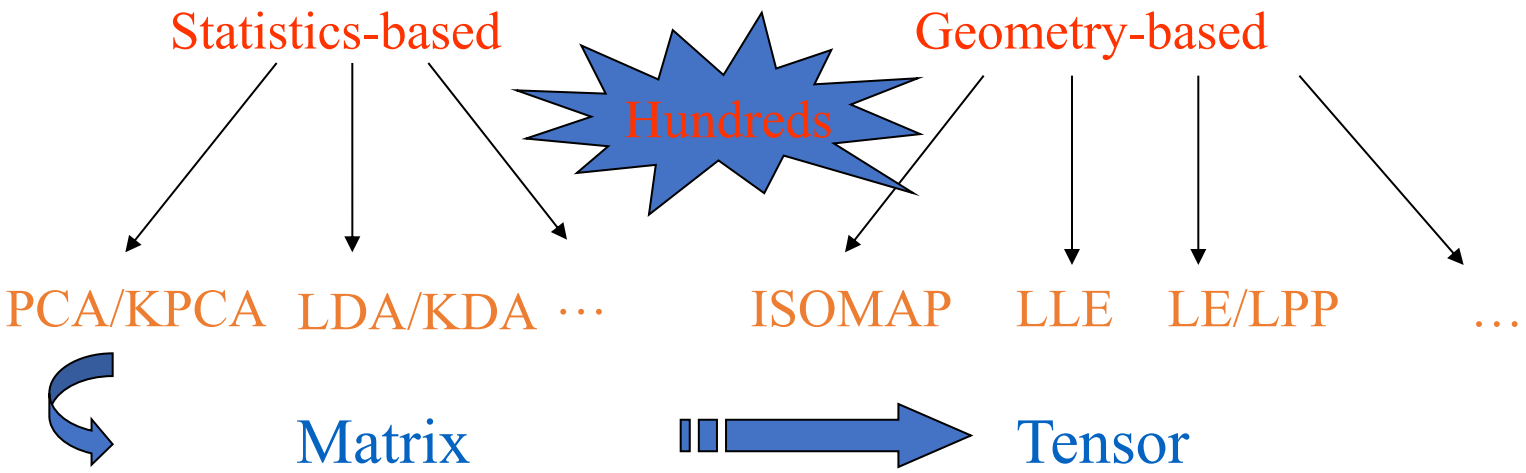
J. Tenenbaum et al., 2000

LE/LPP: Local Similarity Preserving, M. Belkin, P. Niyogi et al., 2001, 2003

LLE: Local Neighborhood Relationship Preserving

S. Roweis & L. Saul, 2000

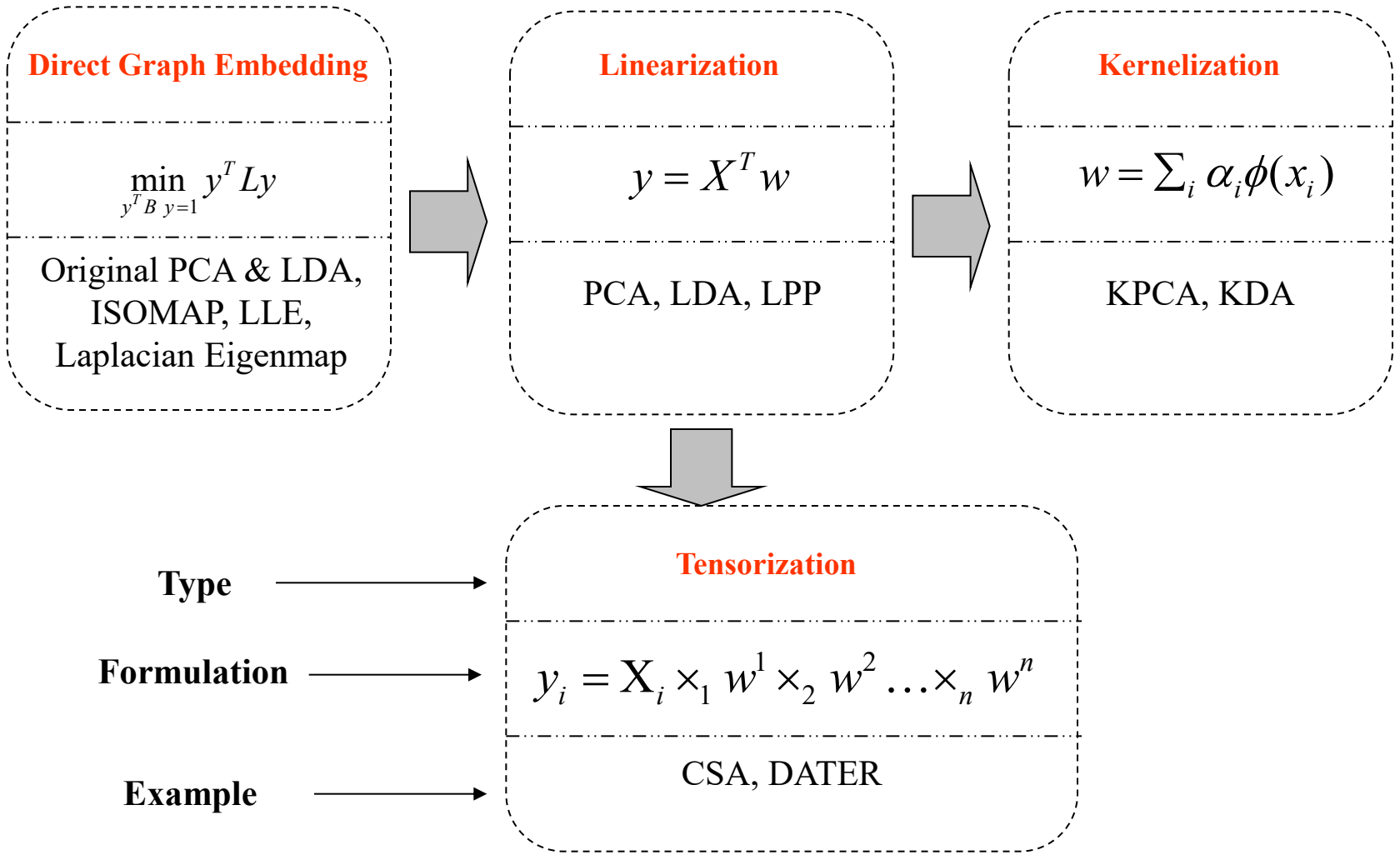
Dimensionality Reduction Algorithms



- Any *common perspective* to understand and explain these dimensionality reduction algorithms? Or any *unified formulation* that is shared by them?

- Any *general tool* to guide developing new algorithms for dimensionality reduction?

Our Answers



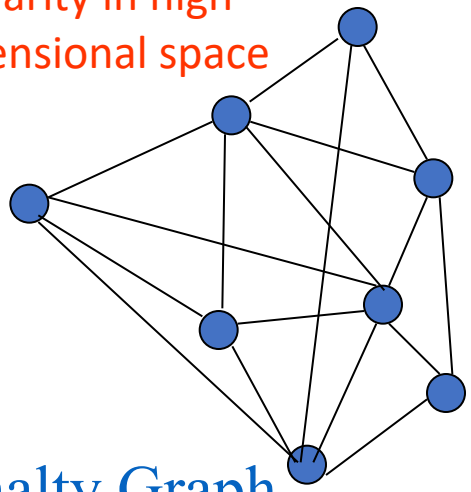
S. Yan, D. Xu, H. Zhang and et al., *CVPR, 2005, T-PAMI,2007*

Direct Graph Embedding

Intrinsic Graph:

$G = [x_i, S_{ij}]$

Similarity in high dimensional space



Penalty Graph

$G^P = [x_i, S_{ij}^P]$

S, S^P: Similarity matrix (graph edge)

L, B: Laplacian matrix from S, S^P;

$L = D - S, \quad D_{ii} = \sum_{j \neq i} S_{ij} \quad \forall i$

Data in high-dimensional space and low-dimensional space (assumed as 1D space here):

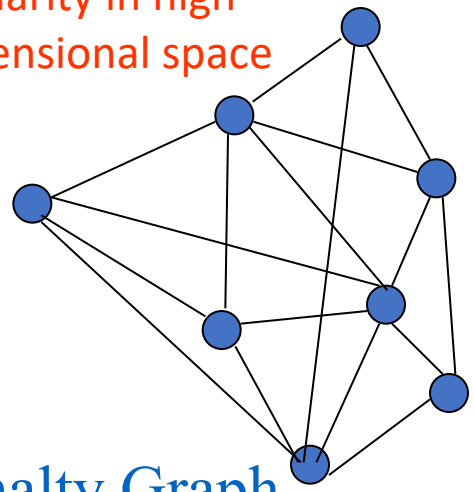
$X = [x_1, x_2, ..., x_N] \quad y = [y_1, y_2, ..., y_N]^T$

Direct Graph Embedding -- Continued

Intrinsic Graph:

$G = [x_i, S_{ij}]$

Similarity in high dimensional space



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L, B: *Laplacian matrix from S, S^P;*

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Data in high-dimensional space and low-dimensional space (assumed as 1D space here):

$X = [x_1, x_2, ..., x_N] \quad y = [y_1, y_2, ..., y_N]^T$

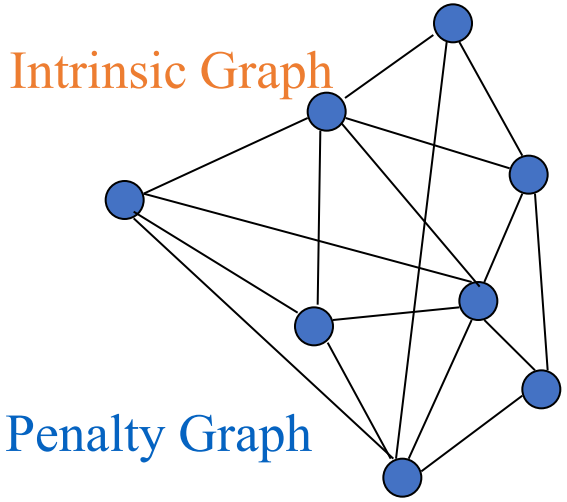
Criterion to Preserve Graph Similarity:

$$y^* = \arg\min \sum_{i \neq j} \| y_i - y_j \|^2 S_{ij} = \arg\min y^T L y$$

$$\arg\min_{y^T y=1 \text{ or } y^T B y=1} \sum_{i \neq j} \| y_i - y_j \|^2 S_{ij} = \arg\min_{y^T y=1 \text{ or } y^T B y=1} y^T L y$$

Problem: It cannot handle new test data.

Linearization



Linear mapping function

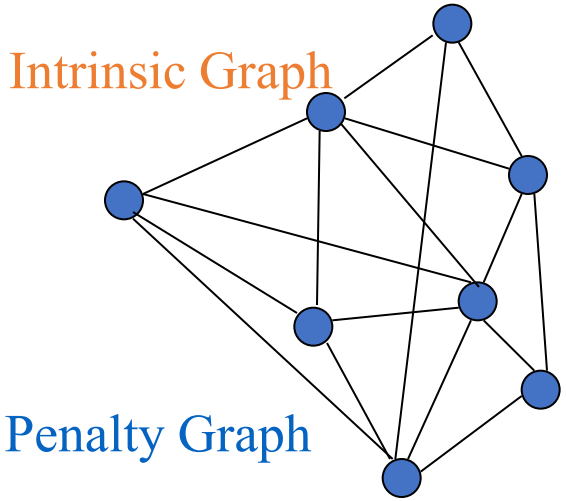
$$y = X^{\mathrm{T}} w$$

Objective function in Linearization

$$w^{*} = \underset{\substack{w^{\mathrm{T}} w=1 \text{ or} \\ w^{\mathrm{T}} XBX^{\mathrm{T}} w=1}}{\arg\min}\ w^{\mathrm{T}} XLX^{\mathrm{T}} w$$

Problem: linear mapping function is not enough to preserve the real nonlinear structure?

Kernelization



Nonlinear mapping: $\phi : x_i \rightarrow \mathcal{F}$

the original input space to another higher dimensional Hilbert space.

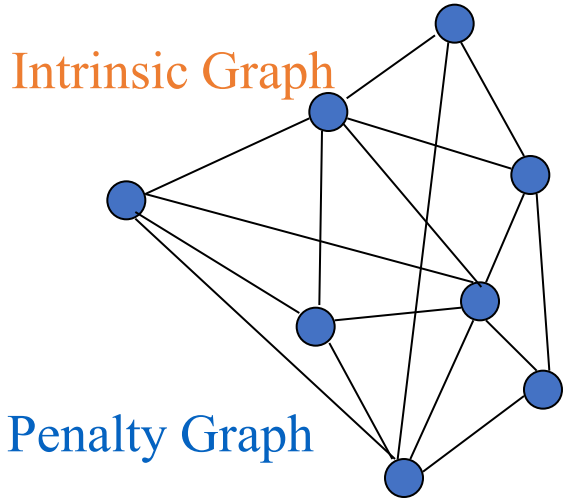
Constraint: $w = \sum_i \alpha_i \phi(x_i)$

Kernel matrix: $K_{ij} = k(x_i, x_j)$ $k(x, y) = \phi(x) \cdot \phi(y)$

Objective function in Kernelization

$$a^* = \arg\min_{\substack{\alpha^T K \alpha = 1 \text{ or} \\ \alpha^T KBK\alpha = 1}} \alpha^T KLK\alpha$$

Tensorization



Low dimensional representation is obtained as:

$$y_i = \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n$$

Objective function in Tensorization

$$(w^1, \ldots, w^n)^* = \underset{f(w^1,\ldots,w^n)=1}{\arg\min} \sum_{i \neq j} \| \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n - \mathbf{X}_j \times_1 w^1 \times_2 w^2 \ldots \times_n w^n \|^2 S_{ij}$$

where

$$f(w^1, \ldots, w^n) = \sum_{i=1}^{N} \| \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n \|^2 B_{ii} \qquad or$$

$$f(w^1, \ldots, w^n) = \sum_{i \neq j} \| \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n - \mathbf{X}_j \times_1 w^1 \times_2 w^2 \ldots \times_n w^n \|^2 S_{ij}^P$$

Common Formulation

Direct Graph Embedding $y^* = \arg\min_{y^T y=1 \text{ or } y^T B y=1} y^T L y$

Linearization $w^* = \arg\min_{w^T w=1 \text{ or } w^T X B X^T w=1} w^T X L X^T w$

Kernelization $a^* = \arg\min_{\alpha^T K \alpha=1 \text{ or } \alpha^T K B K \alpha=1} \alpha^T K L K \alpha$

Tensorization

$$(w^1,\ldots,w^n)^* = \arg\min_{f(w^1,\ldots,w^n)=1} \sum_{i\neq j} \| \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n - \mathbf{X}_j \times_1 w^1 \times_2 w^2 \ldots \times_n w^n \|^2 S_{ij}$$

where $f(w^1,\ldots,w^n) = \sum_{i=1}^{N} \| \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n \|^2 B_{ii}$ or

$f(w^1,\ldots,w^n) = \sum_{i\neq j} \| \mathbf{X}_i \times_1 w^1 \times_2 w^2 \ldots \times_n w^n - \mathbf{X}_j \times_1 w^1 \times_2 w^2 \ldots \times_n w^n \|^2 S_{ij}^P$

A General Framework for Dimensionality Reduction

Algorithm	S & B Definition	Embedding Type
PCA/KPCA/CSA	$S_{ij} = 1/N,\ i \neq j; B = I$	L/K/T
LDA/KDA/DATER	$S_{ij} = \delta_{l_i,l_j} / n_{l_i},\ B = I - \frac{1}{N} ee^{\mathrm{T}}$	L/K/T
ISOMAP	$S_{ij} = \tau(D_G)_{ij},\ i \neq j;\ B = I$	D
LLE	$S = M + M^{\mathrm{T}} - M^{\mathrm{T}} M;\ B = I$	D
LE/LPP	$S_{ij} = \exp\{-\| x_i - x_j \|^2 / t\}$ $if\ \| x_i - x_j \| < \varepsilon;\ B=D$	D/L

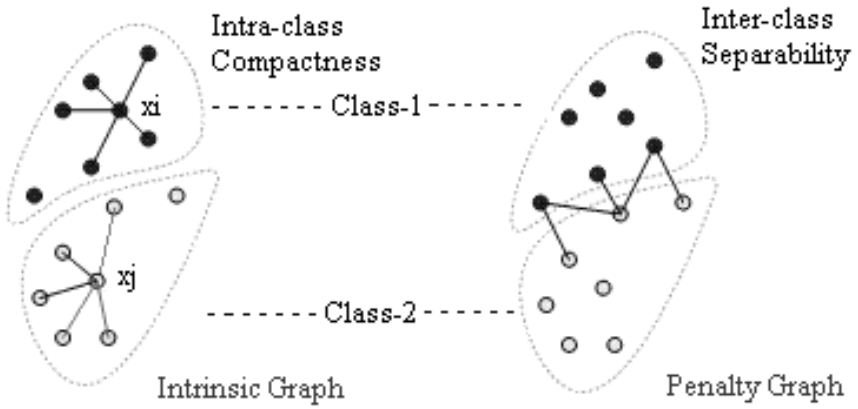
***D*: Direct Graph Embedding**

***L*: Linearization**

***K*: Kernelization**

***T*: Tensorization**

New Dimensionality Reduction Algorithm: *Marginal Fisher Analysis*



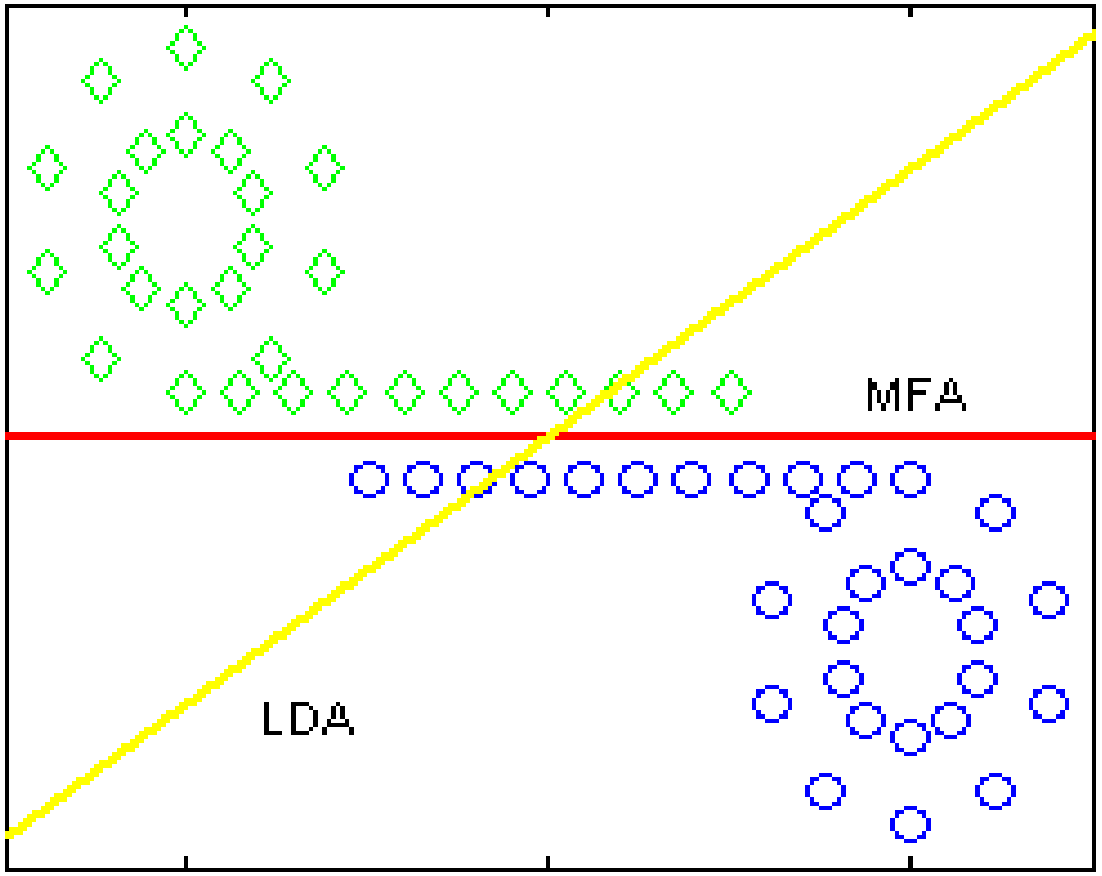
Important Information for face recognition:

1) Label information
2) Local manifold structure
(*neighborhood* or *margin*)

S_{ij} = *1*: *if* x_i is among the k_1-nearest neighbors of x_j in the same class;
0 : *otherwise*

S_{ij}^{P} = *1*: *if* the pair (i,j) is among the k_2 shortest pairs among the data set;
0: *otherwise*

Marginal Fisher Analysis: Advantage



No Gaussian distribution assumption

Experiments: *Face Recognition*

ORL	G3/P7	G4/P6
PCA+LDA (Linearization)	87.9%	88.3%
PCA+MFA (Ours)	89.3%	91.3%
KDA (Kernelization)	87.5%	91.7%
KMFA (Ours)	88.6%	93.8%
DATER-2 (Tensorization)	89.3%	92.0%
TMFA-2 (Ours)	95.0%	96.3%

PIE-1	G3/P7	G4/P6
PCA+LDA (Linearization)	65.8%	80.2%
PCA+MFA (Ours)	71.0%	84.9%
KDA (Kernelization)	70.0%	81.0%
KMFA (Ours)	72.3%	85.2%
DATER-2 (Tensorization)	80.0%	82.3%
TMFA-2 (Ours)	82.1%	85.2%

Summary

- Optimization framework that unifies previous dimensionality reduction algorithms as special cases.

- A new dimensionality reduction algorithm: Marginal Fisher Analysis.

Thank you!